(1) Revit Coordinate Extraction and Tekla Coordination.

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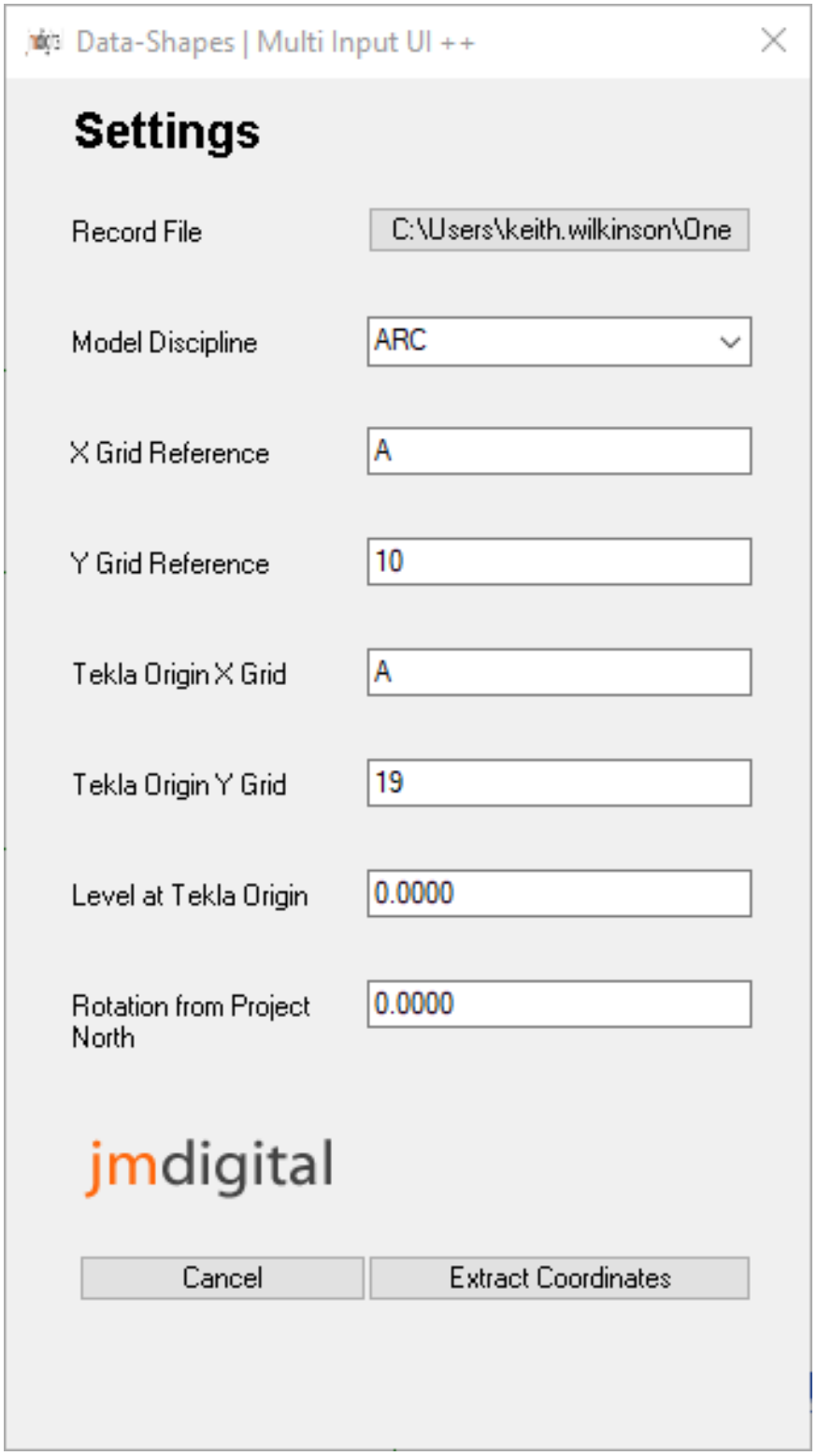
Keith Wilkinson September 16, 2020

Back in 2018, I wrote an article about how to get an IFC out of Tekla that would coordinate correctly with Revit that you can find here: Coordinating between Revit and Tekla using IFC. That article defines the principles that are used within the Dynamo Script discussed below. The script itself automates the process of gathering the necessary numbers to get the correct basepoint translations for correct alignment in both Revit and in other IFC viewers such as Dalux. I'm not going to get into what happens in the graph itself, just the inputs and what they do. Everything in the script exports to an Excel spreadsheet which is also linked at the bottom of this article.

Before getting into the detail of this I'd just like to point something out. If you are linking IFC's into Revit it uses the internal origin point to locate them. This will mean if each discipline's model has a different internal origin point then they will need a separate export from Tekla to get automatic alignment. Just to be clear, this is not Revit's Project Base Point (PBP), that is something different which can be easily adjusted, the internal origin can not.

Extracting the Coordinates

When you run the script you will be faced with the Data-Shapes input form shown below.



Start with choosing the file in which you want to record your coordinates. I've provided a template file below to assist with this but feel free to make your own.

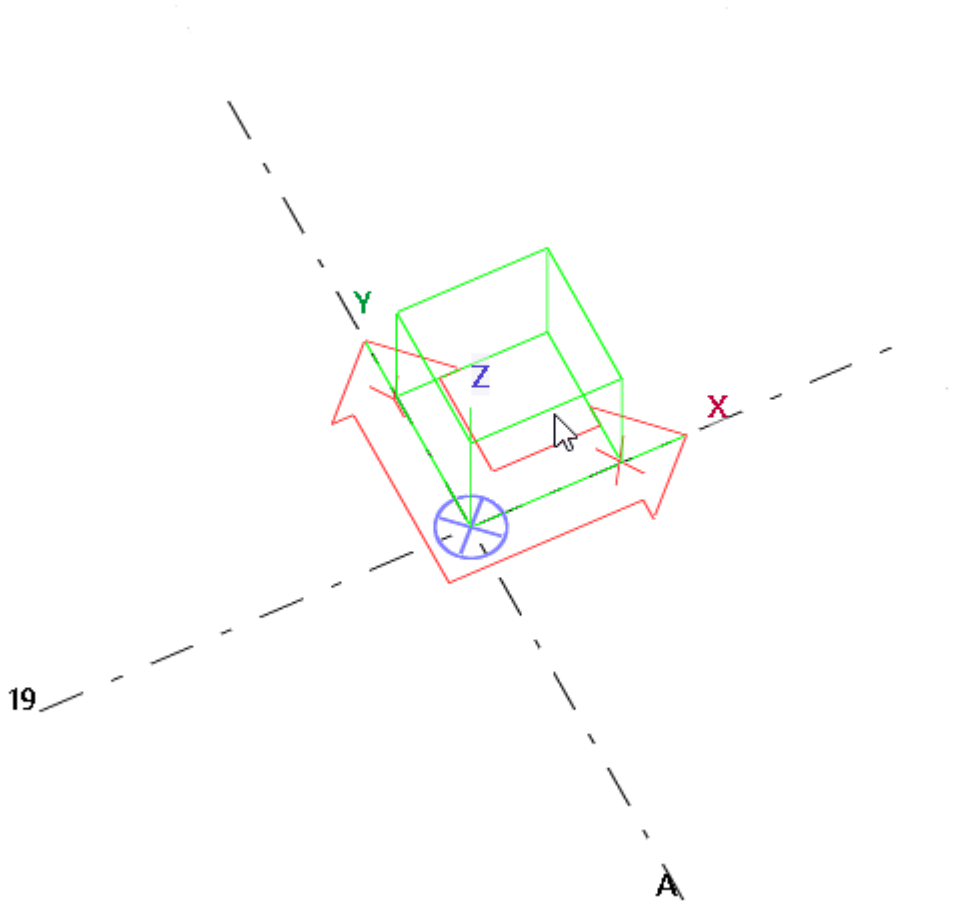
Next, choose what discipline the model relates to. The following are included as standard.

- Architecture (ARC)
- Structure (STR)
- Building Services (MEP)
- Mechanical Services (MEC)
- Electrical Services (ELE)

- Plumbing Services (PLU)
- Furniture (FFE)
- Landscape (LAN)

The next two boxes relate to any given grid intersection within the model. This is just an additional reference point to compare so can be any intersection you like. Make sure the values you enter are an exact match to what is used in the model.

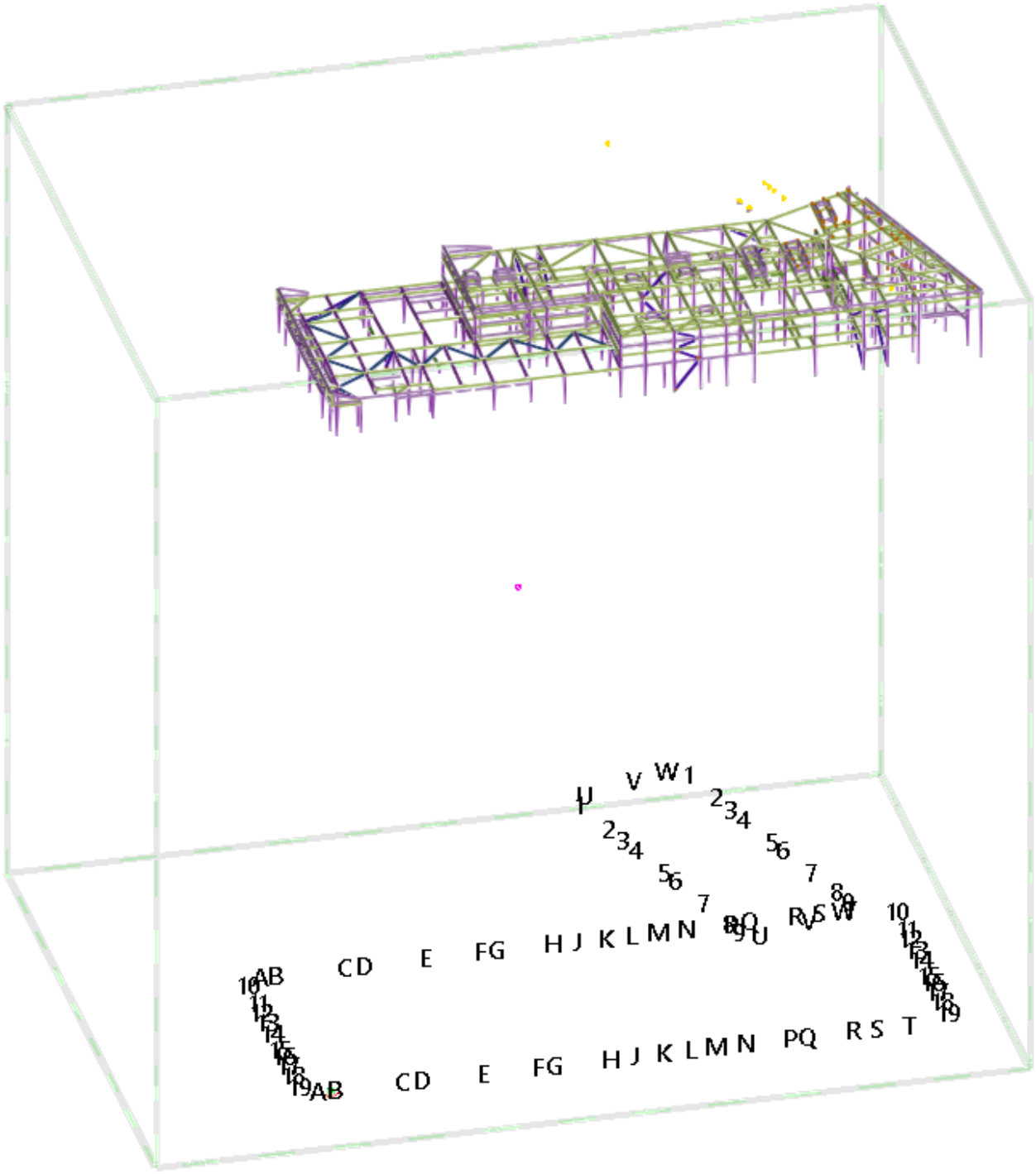
After this, we're into the information relating specifically to the Tekla file. You will need to request this directly from the subcontractor. The first thing we need is the location of the Tekla origin. This is marked by a little green box like the image below.



This point needs to be on a grid intersection (which in fairness it generally seems to be) if it's not the script won't work. Enter the X and Y references as required.

The next bit is the AOD (Above Ordnance Datum) Level of the Tekla base point. In the image below the steelwork has been modeled at the correct AOD for the building so the base point sits at level 0.0 and it is this level that entered into the dialogue box.

Note This is based on the Revit Origin being situated at project level 0.00. It may still work with variations but it's not been tested at the time of writing.



The final bit of information that is required is the rotation of the model from Revit's project North. Now in most instances, this *should* be zero but we have seen models where the model has been rotated 90deg out of alignment. This field caters to that rotation.

Once you have all the details in hit the 'Extract Coordinates' button.

What you get

When you've run the script the following information is extracted and recorded.

Last Updated	17/09/2020	Revision	P01	File Revision	P01

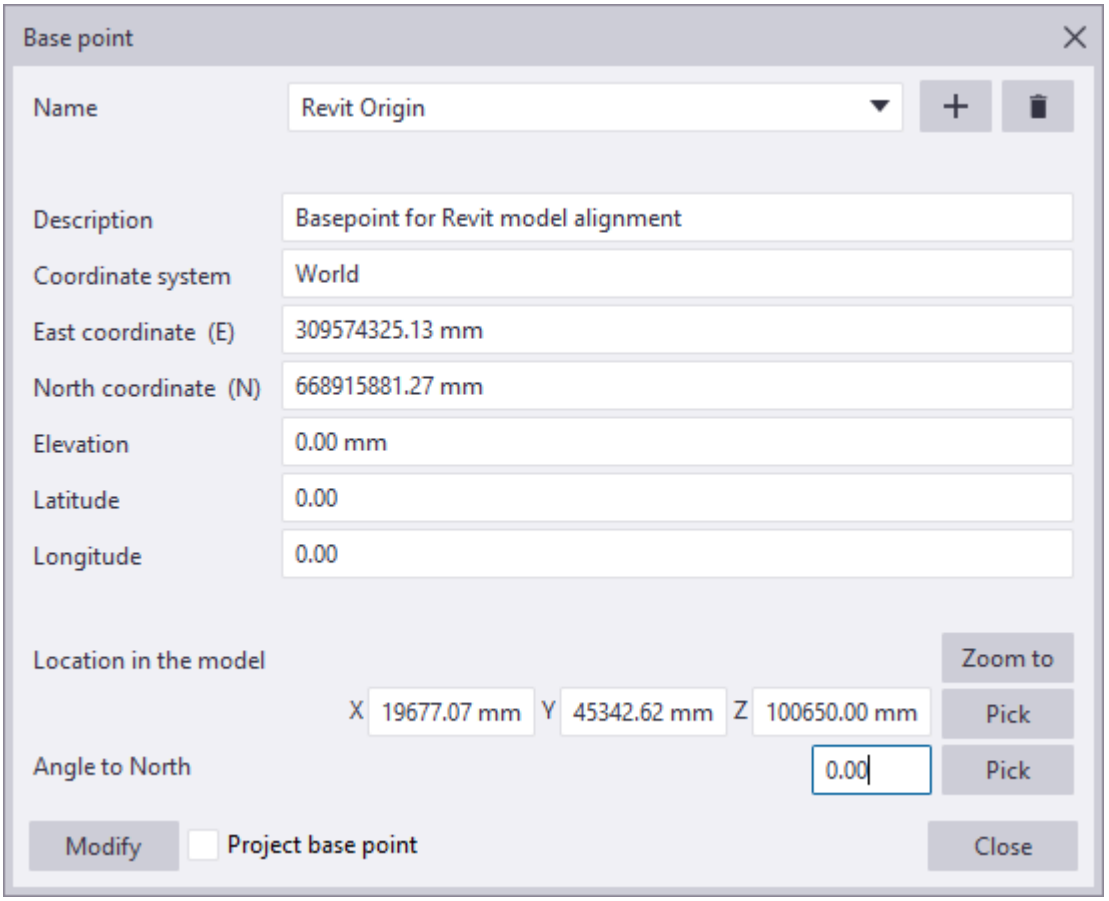
Point	Position	Eastings	Northings	Elevation	Angle to True North
Model Origin	N/A	309574325.1	668915881.3	100650.0	N/A
Project Base Point	N/A	309574325.1	668915881.3	100650.0	22.286488
Survey Point	N/A	0.0	0.0	0.0	N/A
Grid Intersection	A/10	309562332.8	668893255.5	N/A	N/A

	Tekla Origin	A/19				
	Base Point		Location in the Model			
Tekla Settings	East (E)	North (N)	X Offset	Y Offset	Z Offset	Angle To North
Revit	309574325.1297	668915881.2674	19677.0689	45342.6188	100650.0000	0.0000
Dalux	309574325.1297	668915881.2674	19677.0689	45342.6188	0.0000	337.7135

Levels	Level Name	Site FFL	Project FFL	Notes
Finished Floor Levels	Level 00	100650.00	0.00	
	Level 01	104550.00	3900.00	
	Level 02	108570.00	7920.00	
	Level B1	97650.00	-3000.00	

As you can see there is a lot of information gathered. The first section deals with the Revit base points and origins and reports the survey coordinates for these distinct points. These coordinates are valuable to check the alignment of the different models in the project and the spreadsheet is set up to identify variations from the ARC model. Things to watch for are differences in the Model origins and the Grid Intersection.

The next section is what you need to input into Tekla. Within Tekla, you need to define a new basepoint and if you are exporting to Revit and Dalux then you will need at least two. Go to *Menu* > *Project Properties* > *Basepoint* and create a new basepoint inserting the appropriate numbers from your spreadsheet.



Then close and export your IFC using this basepoint, all being well it will align perfectly.

Additional Data

The script also records the elevation, both project and survey, for any levels defined as building stor(e)y. Again this is useful as a check to make sure models are aligned correctly. Lastly, it will record the document revisions, the summary sheet will show the revisions to that tab and also the document revision - the user who ran the script is also recorded along with the data on the Revision tracker tab. There is also a record sheet for each discipline that keeps a history of all recorded data should the comparison be required.

The Files

If you've read this far, thank-you. I hope you will find all this of use, it's something that has plagued me over the last few years. No doubt there will be some scenarios that arise that the script, in its current form, doesn't cope with but if you come across any please let me know and I can look to update it. I think the most likely cause of problems will be to do with elevation but we'll see. Anyway here are the links to the files.

Coordinates Report - Excel Template

Coordinates Report - Dynamo Script

Thanks again.

K.